Calcurion Maths Worksheets

GCSE Mathematics

Topic: Angles & Polygons

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Find the sum of the interior angles of a triangle. **[1]**

2. Find the sum of the interior angles of a quadrilateral. **[1]**

3. Find the sum of the interior angles of a pentagon. **[2]**

4. Find the sum of the interior angles of a hexagon. **[2]**

5. Find the exterior angle of a regular pentagon. **[2]**

6. Find the interior angle of a regular hexagon. **[2]**

7. Find the exterior angle of a regular octagon. **[2]**

8. Find the interior angle of a regular decagon. **[3]**

9. Two angles in a triangle are 50° and 70°. Find the third angle. **[2]**

10. Two angles in a triangle are x° and $2x^\circ$. The third is 60°. Find x. **[3]**

11. Corresponding angles are ..... **[1]**

12. Alternate angles are ..... **[1]**

13. Co-interior (allied) angles sum to ..... **[1]**

14. Find the size of each exterior angle of a regular polygon with 9 sides. **[2]**

15. A regular polygon has exterior angle 24°. How many sides does it have? **[2]**

Section B – Hard (Q16–25)

16. Find the sum of the interior angles of a decagon. **[2]**

17. Find the interior angle of a regular 15-sided polygon. **[3]**

18. The interior angle of a regular polygon is 150°. How many sides does it have? **[3]**

19. In a parallelogram, one angle is 70°. Find the other three angles. **[3]**

20. In an isosceles triangle, the base angles are equal and the vertex angle is 40°. Find the base angles. **[3]**

21. Parallel lines are cut by a transversal. One angle is 65°. Find the corresponding and alternate angles. **[3]**

22. The sum of the interior angles of a polygon is 1260°. How many sides does it have? **[3]**

23. A regular polygon has interior angle 162°. Find the number of sides. **[3]**

24. In a triangle, the angles are in the ratio 2 : 3 : 4. Find the angles. **[4]**

25. Prove that the exterior angles of any polygon sum to 360°. **[4]**

End of Worksheet • Show all working clearly.